

Distinct Principal Spectra of Equal Even-Block Type-A Parabolics

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Abstract

For integers $k, r \geq 1$, consider the type-A parabolic subalgebra of $\mathfrak{sl}_{2kr+1}(\mathbb{C})$ with block sizes $(2k, \dots, 2k, 1)$, where $2k$ occurs r times. We give an explicit all-parameter traversal of its meander, proving that the meander is a single spanning path and hence that the parabolic is Frobenius. Closed formulas for the oriented path potentials then reduce the allowed adjoint eigenvalues to differences of three integer intervals. It follows that the distinct spectrum of the principal element associated with every Frobenius functional is $[-2k+1, 2k] \cap \mathbb{Z}$ for odd r and $[-k, k+1] \cap \mathbb{Z}$ for even r . No assertion about eigenvalue multiplicities or their unimodality is made.

1 Statement and context

Put $d = 2kr + 1$, and partition $[d]$ into r consecutive blocks of size $2k$ followed by a singleton. We write

$$\mathfrak{g}_{k,r} = \mathfrak{p}^A((2k)^r, 1 \mid d) = \{X \in \mathfrak{sl}_d(\mathbb{C}) : X \text{ is upper triangular by these blocks}\}.$$

Thus an off-diagonal matrix unit E_{ij} belongs to $\mathfrak{g}_{k,r}$ precisely when the block containing i is no later than the block containing j . For $F \in \mathfrak{g}_{k,r}^*$ let $B_F(X, Y) = F([X, Y])$. When B_F is nondegenerate, its principal element H_F is characterized by

$$F([H_F, X]) = F(X) \quad (X \in \mathfrak{g}_{k,r}).$$

The index is $\text{ind } \mathfrak{g}_{k,r} = \min_F \dim \ker B_F$, so $\mathfrak{g}_{k,r}$ is Frobenius exactly when this index is zero.

The arXiv v1 of Mayers and Russoniello [3] formulates the support formula below as the support part of its Conjecture 71, together with a stronger multiplicity-unimodality assertion. That source also treats the one-block boundary and other special families. The contribution here is a single spanning-path construction and a potential calculation valid for all k, r . The multiplicities, unimodality, log-concavity, and extended spectrum are outside our statement. This comparison refers specifically to the cited 2023 arXiv version; we make no claim here about present-day priority or about the status of the statement in later versions.

For broad methodological context, exact parameter-dependent spectral bounds also arise in a different operator-theoretic setting in [5]. That work supplies no seaweed-algebra result used here; the relationship is only the use of explicit reductions to certify spectral endpoints.

Theorem 1. *For every $k, r \geq 1$, the Lie algebra $\mathfrak{g}_{k,r}$ is Frobenius. For every Frobenius functional $F \in \mathfrak{g}_{k,r}^*$,*

$$\text{Spec}_{\text{distinct}}(\text{ad } H_F) = \begin{cases} [-2k+1, 2k] \cap \mathbb{Z}, & r \text{ odd}, \\ [-k, k+1] \cap \mathbb{Z}, & r \text{ even}. \end{cases}$$

2 Meander facts used

We briefly fix the conventions needed in the proof. The type-A meander of $\mathfrak{g}_{k,r}$ has vertices $1, \dots, d$. Inside each block of size $2k$, top arcs pair the first vertex with the last, the second with the penultimate, and so on. Bottom arcs make the analogous pairing in the single block $[d]$. Fixed points do not create arcs. If such a meander has C_0 cycle components and P_0 path components, with isolated vertices counted as paths, then

$$\text{ind } \mathfrak{g}_{k,r} = 2C_0 + P_0 - 1. \quad (1)$$

This is the type-A meander index formula of Dergachev–Kirillov [1], in the form summarized in [3, Corollary 3].

Orient top arcs from right to left and bottom arcs from left to right. When the meander is one path, let $w(P_{ij})$ be the number of edges traversed with their orientation minus the number traversed against it along the unique path from i to j . The standard meander principal-element construction [2, 3] says that, with $h_i = w(P_{i,d})$,

$$H = \sum_{i=1}^d (h_i - \bar{h}) E_{ii}, \quad \bar{h} = \frac{1}{d} \sum_{j=1}^d h_j, \quad (2)$$

is the principal element for a Frobenius edge functional. Consequently, $E_{ij} \in \mathfrak{g}_{k,r}$ has adjoint eigenvalue $h_i - h_j$, while the traceless diagonal subspace has eigenvalue zero. The common trace correction in (2) disappears from every difference. Finally, the principal adjoint spectrum of a Frobenius Lie algebra is independent of the chosen Frobenius functional [4, 2]. We will therefore construct one spanning meander and compute its path weights.

3 An all-parameter spanning path

Write the nonsingleton vertices as

$$v_{q,a} = 2k(q-1) + a \quad (1 \leq q \leq r, 1 \leq a \leq 2k),$$

and denote the last vertex by $D = d$. On these vertices the top reflection is

$$T(v_{q,a}) = v_{q,2k+1-a},$$

whereas the bottom reflection is $B(i) = d + 1 - i$. There is no top arc at D and no bottom arc at the fixed point $c = kr + 1$. Set $S = B \circ T$. Direct substitution gives

$$S(v_{q,a}) = \begin{cases} v_{r-q+1,a+1}, & a < 2k, \\ v_{r-q+2,1}, & a = 2k, q > 1, \\ D, & a = 2k, q = 1. \end{cases} \quad (3)$$

First suppose $r = 2s$. Starting at $c = v_{s+1,1}$, list the kr selectors $x_{\ell,a}$ in increasing lexicographic order of $0 \leq \ell < s$ and $1 \leq a \leq 2k$, where

$$x_{\ell,a} = \begin{cases} v_{s+1+\ell,a}, & a \text{ odd}, \\ v_{s-\ell,a}, & a \text{ even}. \end{cases} \quad (4)$$

Equation (3) sends each selector to the next and sends the last one, $v_{1,2k}$, to D .

Now suppose $r = 2s + 1$. Begin the selector list with

$$v_{s+1,k+1}, v_{s+1,k+2}, \dots, v_{s+1,2k}, \quad (5)$$

and then, for $1 \leq \ell \leq s$, append in increasing order of a the vertices

$$x_{\ell,a} = \begin{cases} v_{s+1+\ell,a}, & a \text{ odd}, \\ v_{s+1-\ell,a}, & a \text{ even}. \end{cases} \quad (6)$$

This again gives kr selectors. Formula (3) sends every selector to its successor and the final $v_{1,2k}$ to D . When $s = 0$, only (5) occurs, and the same assertion follows from the first and third cases of (3).

In either parity, if the selector list is $x_1, \dots, x_{kr}$, form

$$x_1, T(x_1), x_2, T(x_2), \dots, x_{kr}, T(x_{kr}), D. \quad (7)$$

Each pair $x_j, T(x_j)$ is a top arc, and $B(T(x_j)) = S(x_j) = x_{j+1}$ gives the following bottom arc; the last such bottom arc ends at D . The vertices in (7) are all distinct. Indeed, for each block pair symmetric about the center, the selectors take odd coordinates in the right block and even coordinates in the left block, while their T -images supply the complementary parities. For odd r , the middle-block selectors (5) and their T -images partition its $2k$ vertices.

For completeness, a top arc in block q has endpoint sum $4k(q-1) + 2k + 1$, which is odd, whereas every bottom arc has endpoint sum $d + 1 = 2kr + 2$, which is even. Thus no top and bottom arc coincide. The walk (7) visits all d vertices and uses $d - 1$ distinct arcs. There are exactly kr top and kr bottom arcs, also $d - 1$ in total, so (7) is the entire meander. Hence $C_0 = 0$ and $P_0 = 1$; (1) proves that $\mathfrak{g}_{k,r}$ is Frobenius.

4 Closed path potentials

For $1 \leq a \leq 2k$, define

$$R(a) = \min\{a - 1, 2k + 1 - a\}, \quad L(a) = R(a) + 1, \quad (8)$$

and

$$M(a) = \begin{cases} 2a - 1, & a \leq k, \\ 2(2k + 1 - a), & a \geq k + 1. \end{cases} \quad (9)$$

Normalize $h_D = 0$. If $r = 2s$, set

$$h_{v_{q,a}} = \begin{cases} L(a), & q \leq s, \\ R(a), & q \geq s + 1. \end{cases} \quad (10)$$

If $r = 2s + 1$, set

$$h_{v_{q,a}} = \begin{cases} L(a), & q \leq s, \\ M(a), & q = s + 1, \\ R(a), & q \geq s + 2. \end{cases} \quad (11)$$

Lemma 2. *The values in (10) and (11) are the oriented path weights $h_i = w(P_{i,D})$.*

Proof. Because the meander is connected and $h_D = 0$, it suffices to verify $h_u - h_v = 1$ on every directed edge $u \rightarrow v$. For a top edge, let its left coordinate be $a \leq k$; the edge is directed from coordinate $2k + 1 - a$ to a . Directly from (8)–(9),

$$\begin{aligned} R(2k + 1 - a) - R(a) &= 1, \\ L(2k + 1 - a) - L(a) &= 1, \\ M(2k + 1 - a) - M(a) &= 1. \end{aligned} \tag{12}$$

Every bottom edge outside an odd middle block is directed from a vertex with profile L to one with profile R . If the left coordinate is $a \geq 2$, the right coordinate is $2k + 2 - a$, and

$$R(2k + 2 - a) = R(a), \quad L(a) - R(2k + 2 - a) = 1.$$

For $a = 1$, the right endpoint has coordinate 1, or is D for the outermost arc; here $L(1) - R(1) = 1$ and $L(1) - h_D = 1$.

For odd r , the remaining bottom arcs start in the middle block. Coordinate 1 maps to coordinate 1 in the first right block, or to D when $r = 1$, and $M(1) - R(1) = M(1) - h_D = 1$. For $2 \leq a \leq k$, the bottom edge remains in the middle block and ends at coordinate $2k + 2 - a$; equation (9) gives

$$M(a) - M(2k + 2 - a) = (2a - 1) - (2a - 2) = 1.$$

These are all directed arcs, proving the lemma. $\square$

The value sets of the profiles are the complete integer intervals

$$L([2k]) = [1, k + 1] \cap \mathbb{Z}, \quad R([2k]) = [0, k] \cap \mathbb{Z}, \quad M([2k]) = [1, 2k] \cap \mathbb{Z}. \tag{13}$$

The last identity follows because the first branch of M produces the odd values and the second branch the even values in this interval.

5 Allowed differences and completion of the proof

An allowed off-diagonal E_{ij} either has i, j in the same block or points from an earlier block to a later block. Its eigenvalue is $h_i - h_j$.

Suppose first that $r = 2s$. Differences within a block, together with the diagonal eigenvalue zero, give every integer in $[-k, k]$. The only mixed profile direction is from L to R , whose differences lie in

$$[1, k + 1] - [0, k] = [1 - k, k + 1].$$

Differences between blocks carrying the same profile and differences from a block to D also lie in $[-k, k + 1]$. Thus no eigenvalue lies outside that interval. The value $k + 1$ occurs from a vertex of potential $k + 1$ to D ; combined with $[-k, k]$, this proves

$$\text{Spec}_{\text{distinct}}(\text{ad } H) = [-k, k + 1] \cap \mathbb{Z}.$$

Now let $r = 2s + 1$. Within the middle block, nonzero differences of the complete interval $[1, 2k]$ give every nonzero integer in $[-2k + 1, 2k - 1]$, and the diagonal part supplies zero. The possible ordered profile comparisons imposed by upper block triangularity satisfy

$$\begin{aligned} L - M &\subseteq [1 - 2k, k], & M - R &\subseteq [1 - k, 2k], \\ L - R &\subseteq [1 - k, k + 1], & L - L, R - R &\subseteq [-k, k]. \end{aligned}$$

The block-to- D differences are values of L , M , or R , hence also lie in $[-2k + 1, 2k]$. There is no allowed right-to-middle or middle-to-left block direction. Therefore there are no eigenvalues outside $[-2k + 1, 2k]$. A middle-block vertex of potential $2k$ paired with D realizes the remaining endpoint $2k$, so

$$\text{Spec}_{\text{distinct}}(\text{ad } H) = [-2k + 1, 2k] \cap \mathbb{Z}.$$

The element H is the principal element of one Frobenius functional. By the functional-independence of the principal adjoint spectrum recalled after (2), both formulas hold for H_F for every Frobenius functional F . This completes the proof of Theorem 1.

Computational and AI assistance disclosure

The proof above is symbolic and does not depend on finite extrapolation. During discovery and checking, exact integer programs reconstructed labelled meander arcs, solved the path potentials by graph traversal, enumerated the allowed matrix units on bounded parameter grids, and tested selected Kirillov matrices over a finite field. These computations served as regression checks only. A generative AI system assisted with exploring the selector formulas, organizing the referee-style checks, and drafting the LaTeX exposition. The author reviewed the final mathematical argument and is responsible for its content.

References

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